SEQUENTIAL COMPOSITION OF POST-QUANTUM ARGUMENTS OF KNOWLEDGE

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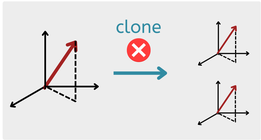
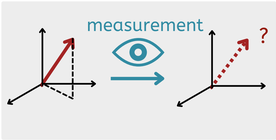
Introduction

Modern digital systems rely on **chaining many verification steps** together, but **quantum computers** threaten to break the assumptions that makes this safe. Current work has carefully developed quantum-secure protocols, but it is **unclear** how **security** and **successfulness** will behave when sequentially composing them.

The **fragility** of quantum states has made analysis a challenge. **No-cloning theorem** and the **measurement effect** means states cannot be copied, and measurements can cause irreversible and difficult to predict perturbations.





With quantum computers being developed by the day, a general framework is needed to ensure that protocols, which underlie areas like **banking**, **computing**, and **messaging,** remain safe before a data apocalypse occurs.

Background

(Informal)

Protocol

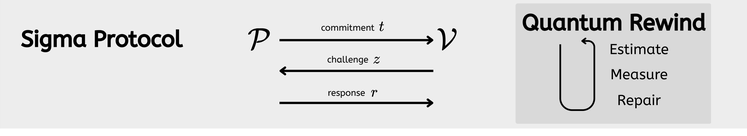
A protocol is a set of rules that specifies how participants exchange information step by step to achieve a goal, such as verifying identity or completing a secure transaction.

Argument of Knowledge (AoK)

An argument of knowledge is a cryptographic protocol where someone (a **prover**) proves to another (a **verifier**) they possess certain information, like a password or solution, without showing the information itself. The key guarantee is that if they can successfully convince you, they must genuinely "know" the secret, not just be guessing or cheating. This allows systems to verify correctness while preserving privacy.

Extractor

An extractor is a hypothetical algorithm used in security proofs that interacts with a prover and pulls out the hidden secret (the witness) if the prover can consistently convince the verifier. In other words, if someone can successfully prove they know something, the extractor shows they must actually know it by recovering the knowledge itself.



Research Work

Technical Goal

- Is there a **general sufficient condition** under which secure/successful sequential composition holds for post-quantum AoKs?
- What does the **knowledge-error loss** look like after sequential composition, and can we minimize it?
- **What protocols** can be sequentially composed while preserving knowledge soundness in the quantum setting?

Research Developments

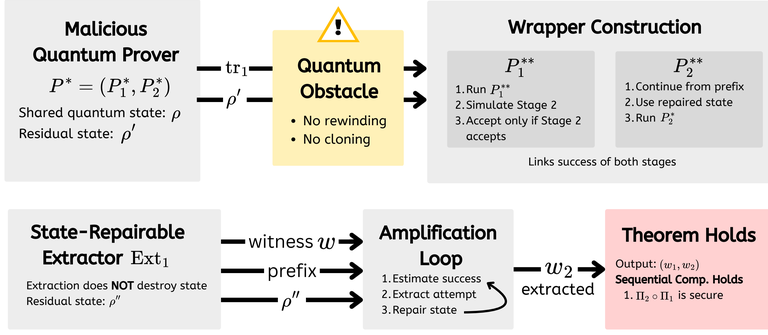
State Repairable Extraction

A quantum protocol Π is *state-repairable* if an extractor can interact with a prover to achieve two things simultaneously:

1. **Witness Extraction**
   The extractor successfully learns the secret (the witness) that the prover is using.
2. **State-Repair**
   After the secret is learned, the prover's quantum state is "fixed" so that it remains just as likely to succeed in the next step of the protocol as it was before the audit.

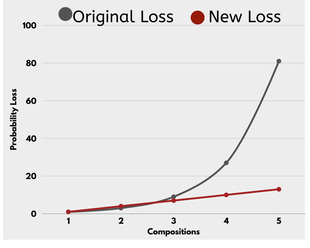
Proof Pipeline

If extraction repairs the prover's quantum state, then we can extract witnesses stage-by-stage, enabling sequential composition.



Success Probability Analysis

Because our amplification loop runs repairs, we create an invariant. This allows us to use union bound instead of multiplying independent error events. This means knowledge-error becomes **linear** and not exponential or quadratic.



Conclusion

- The new notion **state repairable extraction** helps classify the properties required for AoKs to be secure and successful under sequential composition.
- Careful handling of probability of collapse in quantum states helped **decrease knowledge-error growth** from exponential to linear.
- Almost **all current post-quantum protocols** satisfy state repairable extraction, and thus remain secure and successful while chained.

Analysis

Our work shows security protocols can be safely chained together, remain trustworthy and successful even against quantum-powered attackers.

This creates a scalable foundation for future technologies like **secure banking**, **identify verification**, **cloud computing**, **privacy tools**, and **next-generation blockchain** systems which rely on chaining cryptographic protocols.

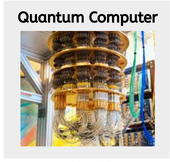


Image courtesy of Google Quantum AI (https://quantumai.google/discover/whatisqc).

Future Work

Future work is necessary to dive into tightening the knowledge-loss, analyzing adaptive composition of AoKs, and streamlining the security proof pipeline of arbitrary quantum protocols. These would all help make quantum protocols more sound, secure, and deployable.

Key Sources & Acknowledgements

[1] M. Bellare and O. Goldreich. On defining proofs of knowledge. In CRYPTO '92, LNCS 740, pp. 390–420. Springer, 1993.
[2] O. Goldreich, S. Micali, and A. Wigderson. Proofs that yield nothing but their validity or all languages in NP have zero-knowledge proof systems. Journal of the ACM, 38(3):691–729, 1991.
[3] A. Chiesa, F. Ma, N. Spooner, and M. Zhandry. Post-quantum succinct arguments: Breaking the quantum rewinding barrier. In FOCS 2022, pp. 49–58. IEEE, 2022.
[4] J. Bootle, V. Lyubashevsky, N. K. Nguyen, and G. Seiler. A non-PCP approach to succinct quantum-safe zero-knowledge. In CRYPTO 2020, LNCS 12171, pp. 441–469. Springer, 2020.
[5] T. Attema, R. Cramer, and L. Kohl. A compressed Σ-protocol theory for lattices. In CRYPTO 2024, LNCS, pp. 549–579. Springer, 2024.
[6] F. Liu, M. Li, and T. Zhao. Asymmetric reduction of knowledge, lattice instantiations, and applications to IVC/PCD. Submitted to CRYPTO 2026.